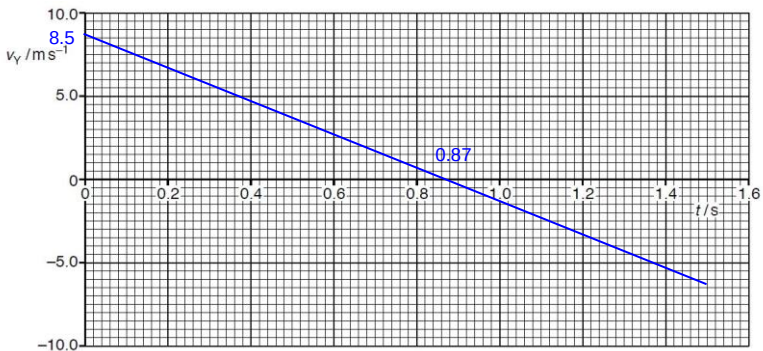


2	
(a)	$v_x = \frac{s_x}{t}$ $= \frac{24}{1.5}$
	$= 16 \text{ m s}^{-1}$
(b)	$\tan 28^\circ = \frac{v_y}{v_x}$ $v_y = (16) \tan 28^\circ$ $= 8.51$
	$= 8.5 \text{ m s}^{-1}$
(c)(i)	
	Straight line with negative gradient
	Gradient $= 9.81 \text{ m s}^{-2} \rightarrow t = 0.87 \text{ s} \rightarrow \text{cut at } 0.84 \text{ s} < t < 0.90 \text{ s}$
	Stop at $t = 1.5 \text{ s}$, magnitude of v_y must be less than initial
(c)(ii)	maximum height = area under the v-t graph
	$= \frac{1}{2}(0.866)(8.5)$ $= 3.7 \text{ m}$
(c)(iii)	$\frac{\text{k.e.}}{\text{g.p.e}} = \frac{\left(\frac{1}{2}\right)mv_x^2}{mgh}$ $= \frac{\left(\frac{1}{2}\right)(16)^2}{(9.81)(3.7)}$
	$\frac{\text{k.e.}}{\text{g.p.e}} = 3.5$
(iv)	Steeper initial gradient, which decreases over time. Area above and below must be the same.
	Tangent to curve where it cuts the time-axis is parallel to original graph and must be the correct shape.

Anglo-Chinese Junior College
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